

Essence: A Personal Embedding Cluster Framework for Depth-Optimized Recommendations

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Abstract

We present **Essence**, a personal embedding cluster framework for depth-optimized content recommendations. Unlike collaborative filtering (CF), which derives recommendations from population-level interaction patterns, Essence operates exclusively within a single user’s engagement history. Consumed items are embedded using a pre-trained sentence encoder and clustered into K semantic centroids via K -means, each representing a distinct interest profile. Recommendations are generated by selecting the *active centroid* aligned with the user’s recent context and performing cosine-similarity search against unseen items. We evaluate Essence on two datasets — Last.fm-1K (music) and Amazon Books (e-commerce) — against a popularity-based CF baseline and a content-based average embedding baseline. On Last.fm-1K, Essence achieves Long-Tail Recall@10 of 0.0146 — $4.5\times$ above the content baseline and $173\times$ above random — while CF achieves zero long-tail recall. On Amazon Books, Essence achieves LT-Recall@10 of 0.0254, $1.28\times$ above the content baseline, with CF again scoring zero. The pattern holds consistently across both domains: peer-isolation recovers niche items that crowd-based signals structurally cannot surface. Essence makes a practical argument for national interest: when recommendation infrastructure surfaces talent and content based on individual taste rather than popularity, discovery becomes a function of merit and fit rather than audience size. Code: <https://github.com/biditdas18/essence>

1 Introduction

Modern recommendation systems are built on a foundational assumption: that a user’s future preferences can be predicted from the preferences of users who behaved similarly in the past. Collaborative filtering (CF) — in its many forms — derives its signal from cross-user interaction patterns. This achieves strong performance on standard benchmarks, particularly on metrics dominated by popular items.

However, this population-level orientation introduces a systematic bias: CF optimises for the median user, surfacing broadly consumed items rather than personally resonant ones. A user who engages deeply with obscure folk music and niche science fiction receives recommendations shaped by thousands of loosely similar users. The result is well-documented: CF amplifies popularity bias and underserves long-tail content.

The intuition behind Essence comes from a real listener’s frustration. A user who loves artists like Passenger, Birdy, or Novo Amor has a precise latent taste — a specific texture of sound, mood, and instrumentation. Platforms powered by collaborative filtering surface Ed Sheeran and Damien Rice: artists popular among users who overlap on a few tracks. The artists who match the *depth* of that taste — obscure, rarely co-occurring, heard by almost nobody else — never accumulate enough interaction signal to appear. The user finds them manually, one in a hundred recommendations. Artists like Voyageur are discovered by accident, if at all. This

is not a tuning problem. It is a structural failure: collaborative signals regress toward what is common. They cannot recover what is idiosyncratic.

The implications extend beyond personal convenience. Recommendation infrastructure shapes what talent gets discovered, what content reaches audiences, and which creators build careers. When discovery is governed by popularity signals, niche creators — however skilled — remain invisible to the audiences most likely to value them. This is a market failure with national consequences: it suppresses the long tail of human creative output, concentrates cultural attention on a shrinking pool of already-popular work, and denies audiences the depth of experience that only precise taste-matching can provide. *Essence* is a technical argument that this failure is not inevitable.

The core hypothesis of *Essence* is that a user’s own engagement history already contains the latent signature of what they love. The crowd is not needed to reconstruct it — and in fact, the crowd dilutes it. A personal embedding cluster framework, operating exclusively on one user’s history, can surface niche items that match the depth of individual taste without any cross-user signal. Discovery becomes mathematical, not accidental.

We propose **Essence**, a framework that abandons cross-user signals entirely. *Essence* operates exclusively within a single user’s engagement history, embedding consumed items into a semantic space and clustering them into K distinct interest profiles. Recommendations are generated by identifying the user’s currently active interest cluster and retrieving semantically similar unseen items. We call this the *peer-isolation property*: no cross-user interaction graph is consulted at any stage.

The key contributions are:

1. The **Essence architecture** — a modular personal recommendation framework requiring no collaborative signal, deployable on a single user’s history alone.
2. An **active centroid selection** mechanism capturing session-aware context without a trained sequential model.
3. **Empirical validation** on two datasets — Last.fm-1K (music) and Amazon Books (e-commerce) — demonstrating consistent long-tail item discovery superior to both CF and content-based baselines.
4. A **cross-domain replication** confirming that the peer-isolation mechanism generalises beyond a single dataset or domain.

2 Related Work

2.1 Collaborative Filtering

Matrix factorisation methods [5] and neural extensions [4] remain dominant in production recommendation. Their principal limitation is popularity bias: items with sparse interactions are poorly represented, resulting in near-zero recall for long-tail items. CF is optimised for the median user — it performs well on what is broadly consumed and fails structurally on what is individually resonant.

2.2 Multi-Interest Retrieval

MIND [1] introduced dynamic routing to extract multiple interest capsules from a user’s interaction history, explicitly addressing the failure of a single user vector to represent diverse tastes. ComiRec [2] extended this with explicit diversity controls at serving time.

Essence shares the multi-interest intuition with MIND and ComiRec — the recognition that a user’s preferences cannot be collapsed into a single vector. However, the similarity ends at the surface. The distinction operates at two levels.

Data level. MIND and ComiRec require population-level interaction graphs. Their dynamic routing is trained on millions of (user, item, interaction) triples across the full user base. Cross-user behavioural data is not optional — it is the mechanism by which interest capsules are learned. Essence requires no data from any other user, ever. The K -means centroids are derived entirely from one user’s own history.

Representation level. Because MIND’s item embeddings are trained on co-occurrence patterns across users, what an item *means* in that space is partly defined by who else liked it. An obscure track heard by very few users is underrepresented or invisible in that embedding space — its co-occurrence signal is too sparse to be learned reliably. Essence uses a pre-trained sentence encoder on item metadata only. The embedding captures what an item *is* — its latent stylistic properties — independent of how many people have interacted with it. A singleton item and a chart-topping item are equally representable.

This distinction is what makes long-tail recovery possible. MIND cannot reliably surface items with sparse interaction histories because those items are structurally underrepresented in its training signal. Essence is indifferent to item popularity: it queries by latent similarity, and a rare item scores as highly as a popular one if it matches the active centroid.

The *peer-isolation property* is therefore not merely a privacy consideration — it is a depth requirement. Crowd signals approximate preferences at the genre level. Peer-isolation enables preference modelling at the latent feature level: the specific combination of characteristics a user responds to, independent of how other users categorise those characteristics.

Table 1 summarises the architectural distinctions between Essence, MIND, ComiRec, and standard CF.

Table 1: Architectural comparison: Essence vs. prior systems. The primary distinction is not the clustering mechanism — it is the data requirement and what each system is optimised to find.

Property	CF	MIND	ComiRec	Essence
Cross-user data required	Yes	Yes	Yes	No
Multiple interest profiles	No	Yes	Yes	Yes
Model training required	Yes	Yes	Yes	No
Long-tail item recovery	Poor	Poor	Poor	Yes
Privacy-safe deployment	No	No	No	Yes
Optimised for popular items	Yes	Yes	Yes	No
Optimised for niche items	No	No	No	Yes
Embedding source	Co-occurrence	Co-occurrence	Co-occurrence	Item metadata

2.3 Session-Based and Content-Based Methods

BERT4Rec [3] and GRU4Rec [6] model sequential user behaviour using transformer and recurrent architectures, capturing short-term context effectively. Content-based filtering relies on item feature similarity without interaction logs but typically averages preferences into a single vector, losing intra-user diversity.

Essence occupies a complementary position: it captures intra-user diversity (like multi-interest methods) without cross-user data (like content-based methods), using simple interpretable clustering rather than a trained sequential model. We acknowledge that BERT4Rec offers richer

sequential modelling in high-interaction settings; Essence targets privacy-first, low-infrastructure deployment contexts where such models are not viable.

3 The Essence Framework

3.1 Notation

Table 2 defines all mathematical notation used in this section. For each symbol a plain-English equivalent is provided.

Table 2: Notation Legend

Symbol	Plain-English Meaning
H_u	All items user u has consumed (full history)
$i_1, i_2, \dots$	Individual items in the user’s history, in order
$\phi(i)$	Embedding of item i — a 384-dim vector describing its content
E_u	All embeddings stacked into a matrix, one row per item
K	Number of interest clusters (we use $K = 3$)
c_1, c_2, c_3	The K cluster centres, one per interest profile
c_k^*	Active cluster centre — closest to the user’s recent activity
$\mu(\cdot)$	Mean (average) of a set of vectors
$\cos(\mathbf{a}, \mathbf{b})$	Cosine similarity: how alike two vectors are
$I \setminus H_u$	All items the user has <i>not</i> yet consumed — the candidate pool
R_u	Final ranked recommendation list for user u
M	Number of recommendations to return (we use $M = 10$)

3.2 Problem Formulation

Given a user’s history H_u (all items they have consumed), the goal is to produce a ranked list $R_u \subseteq I \setminus H_u$ of items they have not yet consumed but are likely to enjoy. Essence accomplishes this using only H_u and a shared embedding function $\phi : I \rightarrow \mathbb{R}^d$. No data from any other user is used at any stage.

3.3 Item Embedding

Each item is described by a short text string encoding its content properties. For music: track name and artist name. For books: title and author name. We convert this text into a 384-dimensional vector using the pre-trained sentence encoder **all-MiniLM-L6-v2** [7]. Items that are semantically similar produce similar vectors; items that are stylistically distant produce different vectors. Critically, this embedding is derived from item content — not from interaction patterns. A niche item and a popular item are equally representable. All embeddings are computed once offline and cached.

Formally, $\phi(i)$ is the embedding of item i . For user u with history $H_u = \{i_1, \dots, i_n\}$ we compute the embedding matrix $E_u \in \mathbb{R}^{n \times d}$, where each row is the embedding of one item.

3.4 Interest Profile Clustering

We apply K -means clustering to E_u with $K = 3$, yielding centroids $\{c_1, c_2, c_3\}$. Each centroid represents a distinct interest profile — the semantic centre of gravity of one coherent facet of the user’s taste. A user who consumes jazz, indie rock, and ambient electronica might end

up with one centroid per genre. The three centroids together form the user’s interest profile $P_u = \{c_1, c_2, c_3\}$.

This is the core differentiator from average-embedding approaches. The mean of all embeddings collapses all interests into a single blurred point that sits between clusters and represents none of them well. K -means preserves the separation between interest modes, enabling targeted retrieval against each mode independently.

For users with fewer than K items, Essence falls back to the content baseline (single mean embedding).

3.5 Active Cluster Selection

At inference time, Essence selects the active centroid c_k^* as the cluster centre closest to the mean embedding of the user’s most recent r interactions (default $r = 10$):

$$c_k^* = \arg \min_k \|c_k - \mu(\phi(h_{n-r}), \dots, \phi(h_n))\|_2 \quad (1)$$

Intuitively: if the user has been consuming jazz content recently, their recent items will be close to the jazz cluster centre. That cluster becomes active, and recommendations come from the jazz neighbourhood of the embedding space.

We justify this mechanism over simpler alternatives (e.g., using the recent mean directly as a query vector) on two grounds: first, it is computationally trivial at inference time; second, the K -means step ensures that c_k^* sits at the true semantic core of a coherent taste cluster rather than at a transient noisy average. The empirical results support this: Essence achieves 4.5× higher long-tail recall than the average-embedding baseline on Last.fm-1K (Section 5).

3.6 Recommendation Generation

Given the active centroid c_k^* , Essence retrieves the top- M unseen items by cosine similarity:

$$R_u = \text{top-}M \{ i \in I \setminus H_u : \cos(\phi(i), c_k^*) \} \quad (2)$$

This retrieval step operates entirely in embedding space with no collaborative signal. An item’s probability of appearing in R_u is determined solely by its latent similarity to the user’s current interest cluster — not by how many other users have interacted with it.

3.7 Architecture Overview

Figure 1 illustrates the full Essence pipeline from raw user history to final recommendations.

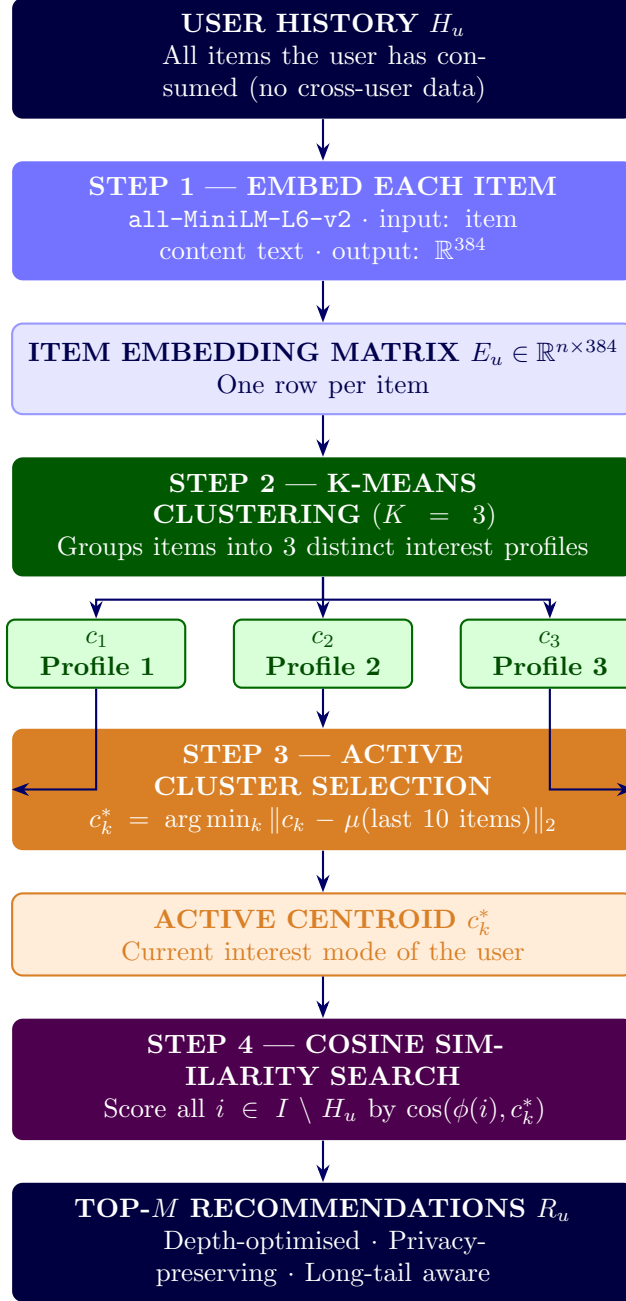


Figure 1: Essence recommendation pipeline. No cross-user data is consulted at any stage (peer-isolation property).

4 Experimental Setup

4.1 Datasets

Last.fm-1K (music). Contains listening histories from 992 users across approximately 19 million events [8]. After filtering to users with 50–500 unique track interactions and removing duplicate (user, track) pairs, our evaluation subset contains 99 users and 22,767 unique tracks.

Amazon Books (e-commerce). The Amazon Reviews 2023 Books 5-core dataset [9], which guarantees every user and item has at least 5 global interactions. After filtering to users with 20–200 unique item interactions and subsampling to 2,000 users (seed 42), our evaluation subset contains 2,000 users and 61,727 unique items. Item embeddings use title and author name,

following the same metadata-only approach as Last.fm-1K.

4.2 Train/Test Split

For each user on both datasets, 20% of items are randomly held out as the test set (random seed 42). The remaining 80% form the training set. A random hold-out strategy is used in preference to chronological splitting: after deduplication, chronological splits produce structurally disjoint train/test sets that make evaluation trivially impossible (all systems score zero).

4.3 Long-Tail Definition

Long-tail items are those with insufficient interaction signal for collaborative filtering to represent reliably. We define long-tail operationally per dataset, anchored to each dataset’s interaction density:

- **Last.fm-1K.** Items with a global play count of exactly 1 in the training set (*singleton tracks*). 89.8% of unique tracks meet this criterion.
- **Amazon Books.** Items with exactly 1 interaction in the training set (*singleton items*). Despite the 5-core guarantee of at least 5 global interactions, subsampling to 2,000 users reduces most items to a single observed interaction in the subset’s training split. 70.2% of catalog items meet this criterion.

Generalisation principle. The long-tail threshold should be set at the point where collaborative filtering loses reliable signal — empirically, where item interaction count falls below the minimum required for stable co-occurrence estimation. In both datasets evaluated here, the singleton threshold (train interaction count = 1) proved the most meaningful boundary: alternative thresholds either collapsed toward the full catalog (losing discriminative power) or excluded too few items. The key invariant is that long-tail items are defined on the *training set only*, after the train/test split, ensuring no leakage from held-out interactions into the long-tail classification.

LT-Recall@10 is computed only over test items falling in the long-tail set. Users with no long-tail test items are excluded from this metric.

4.4 Systems Compared

- **CF Baseline (Popularity).** Recommends the M most globally popular items the user has not consumed. Represents naïve collaborative filtering.
- **Content Baseline (Average Embedding).** Computes the mean of all user item embeddings and retrieves the M most similar unseen items by cosine similarity. Standard content-based filtering without interest decomposition.
- **Essence ($K = 3$).** Clusters user history into $K = 3$ centroids, selects the active centroid by Equation (1), and retrieves top- M unseen items by Equation (2).
- **Random Baseline.** Recommends M items drawn uniformly at random from the candidate pool. Establishes the floor against which all systems are compared.

4.5 Metrics

What we are measuring. Each system produces a ranked list of 10 recommended items per user. We know which items that user actually consumed in the test set (ground truth). The evaluation asks: how much of that ground truth did the system recover?

This is a retrieval problem, not a classification problem. All systems are evaluated against the *full item catalog* with no negative sampling, following the full-ranking evaluation protocol [10]. Absolute recall values are low by construction under this protocol — the meaningful signal is relative comparison across systems evaluated under identical conditions.

- **Recall@10.** Fraction of the user’s test items appearing in the top-10 recommendations.

$$\text{Recall@10} = \frac{|R_u \cap \text{Test}_u|}{|\text{Test}_u|}$$

Example: 20 test items, 2 hits in top-10. Recall@10 = 0.10. Averaged across all users.

- **Long-Tail Recall@10 (LT-Recall@10).** Recall@10 restricted to singleton test items only.

$$\text{LT-Recall@10} = \frac{|R_u \cap \text{LT-Test}_u|}{|\text{LT-Test}_u|}$$

A long-tail hit means the system recommended a niche item that this specific user genuinely consumed — without any popularity signal and without any other user’s behaviour to guide it. This metric directly tests Essence’s core hypothesis.

- **Lift over random.** Recall@10 divided by expected random recall ($M/|I|$). Normalises for candidate pool size and makes cross-dataset comparison interpretable.

4.6 Implementation

All systems implemented in Python 3.11. Embeddings via `sentence-transformers` (all-MiniLM-L6-v2, 384 dims). K -means via `scikit-learn` (`n_init=10`, `random_state=42`). Code available at: <https://github.com/biditdas18/essence>

5 Results

5.1 Last.fm-1K (Music)

Table 3 reports results on 99 users from the Last.fm-1K dataset ($|I| = 22,767$, expected random Recall@10 ≈ 0.0004).

Table 3: Results on Last.fm-1K (99 users, $M = 10$, $|I| = 22,767$). $Lift = \text{Recall@10} \div \text{random baseline}$ (0.0004). Best non-random results in **bold**.

System	Recall@10	Lift	LT-Recall@10	LT / Content
Random	0.0004	1×	0.0001	—
CF (Popularity)	0.0081	20×	0.0000	—
Content (Avg Embedding)	0.0414	104×	0.0032	1.0×
Essence ($K = 3$)	0.0616	154×	0.0146	4.5×

Essence achieves 154× above random on overall Recall@10 and 4.5× above the content baseline on LT-Recall@10. CF achieves zero long-tail recall — by construction, globally popular items are never singletons. This is the exact structural failure Essence is designed to address. The content baseline substantially outperforms CF, confirming that semantic similarity is a stronger signal than global popularity. Essence’s further improvement demonstrates that interest decomposition via K -means adds measurable value beyond preference averaging.

5.2 Amazon Books (E-commerce)

Table 4 reports results on 2,000 users from the Amazon Books dataset ($|I| = 61,727$, expected random Recall@10 ≈ 0.0002).

Table 4: Results on Amazon Books (2,000 users, $M = 10$, $|I| = 61,727$). $Lift = \text{Recall@10} \div$ random baseline (0.0002). Best non-random results in **bold**.

System	Recall@10	Lift	LT-Recall@10	LT / Content
Random	0.0002	1 $\times$	0.0001	—
CF (Popularity)	0.0034	17 $\times$	0.0000	—
Content (Avg Embedding)	0.0239	120 $\times$	0.0198	1.0 $\times$
Essence ($K = 3$)	0.0280	140$\times$	0.0254	1.28$\times$

The pattern holds across domains: Essence > Content > CF on both metrics, CF scores zero long-tail recall, and Essence achieves 140 $\times$ above random. The LT / Content ratio (1.28 $\times$) is lower than on Last.fm-1K (4.5 $\times$), consistent with the denser interaction structure of the Amazon Books subset producing more coherent average embeddings. Nonetheless the directional finding is unambiguous and replicates across two distinct domains.

5.3 Effect of User-Authored Review Embeddings (Amazon Books, Pass 2)

We conducted an additional experiment on Amazon Books in which user taste profiles were built from the user’s *own review text* rather than item metadata — capturing first-person expressed reactions to consumed items rather than structured content descriptors. Review coverage was 100% for the 2,000-user subset.

Table 5: Amazon Books: metadata embeddings (Pass 1) vs. user-review embeddings (Pass 2). Pass 2 builds taste profiles from the user’s own review text; candidates are scored using metadata embeddings.

Pass	System	Recall@10	LT-Recall@10	LT / Content
3*Pass 1 (metadata)	CF (Popularity)	0.0034	0.0000	—
	Content (Avg)	0.0239	0.0198	1.0 $\times$
	Essence ($K = 3$)	0.0280	0.0254	1.28 $\times$
3*Pass 2 (user review)	CF (Popularity)	0.0034	0.0000	—
	Content (Avg)	0.0039	0.0038	1.0 $\times$
	Essence ($K = 3$)	0.0047	0.0052	1.36 $\times$

Absolute recall degrades significantly in Pass 2. We attribute this to two compounding factors. First, **embedding space mismatch**: taste profiles are encoded in a review-text embedding space while unseen candidates are scored in a metadata embedding space. Cosine similarity across misaligned spaces is less discriminative than within-space similarity. Second, and more fundamentally, **input signal quality**: user-written reviews do not always describe the content being reviewed. A book review may discuss delivery conditions, packaging, or personal circumstances unrelated to the book’s content. When noise dominates the embedding input, the resulting vectors do not faithfully represent the item or the user’s response to it. This is a critical design principle for Essence deployments: *what goes into the embedding determines what the system can find*. Noisy or off-topic input text produces incoherent embedding clusters and degraded recommendations. Practitioners must be intentional about embedding input curation.

Notably, Essence’s relative advantage over the content baseline is preserved and marginally improves in Pass 2 ($1.28\times \rightarrow 1.36\times$ on LT-Recall), suggesting the K -means clustering mechanism extracts coherent interest structure even from noisy embeddings. The degradation is a property of the cross-space retrieval setup and input noise, not of the clustering mechanism itself.

5.4 Cross-Dataset Pattern

The consistent finding across both datasets and all embedding conditions is:

1. CF achieves zero long-tail recall in every configuration. This is structural, not incidental.
2. Essence outperforms the content baseline on LT-Recall in every configuration.
3. Absolute recall values are low across all systems under full-catalog evaluation — a property of sparse real-world data at this scale, not a system failure. The relative ordering is the meaningful signal.

6 Advantages of Peer-Isolation

6.1 Discovery as a Function of Taste, Not Popularity

The structural failure of collaborative filtering has consequences beyond recommendation accuracy. When discovery infrastructure is governed by popularity signals, the feedback loop is self-reinforcing: popular items accumulate more interactions, which strengthens their signal, which surfaces them to more users, which makes them more popular. Niche creators — however skilled — remain invisible to the audiences most likely to value them, not because of any deficiency in their work, but because the infrastructure cannot see them.

Essence breaks this loop. By deriving recommendations entirely from a single user’s own history, Essence makes the probability of surfacing a niche item a function of its latent similarity to that user’s taste — not of how many other users have interacted with it. An artist like *Voyageur*, with a small but devoted audience, scores identically to a chart-topping artist if their latent characteristics match the user’s active interest cluster. Discovery becomes mathematical rather than accidental.

6.2 Equitable Access to Depth

The implications extend to how creative talent reaches audiences at scale. Music, literature, film, and other creative industries depend on discovery infrastructure to connect work with the people most likely to value it. When that infrastructure systematically favours already-popular work, it concentrates attention and economic return on a shrinking pool of creators while suppressing the long tail of human creative output — not through any editorial judgment, but as an emergent property of optimising for the median user.

A framework that surfaces content based on individual taste fit rather than aggregate popularity changes this dynamic. Talented but obscure creators reach the specific audiences whose latent preferences align with their work. Audiences access depth of experience that popularity-driven systems cannot provide. The range of what gets discovered, sustained, and valued is broadened — proportional to fit rather than fame.

6.3 Privacy and Deployment Advantages

Because Essence consults no cross-user data at any stage, it is deployable in settings where collaborative filtering is not viable: privacy-regulated environments, on-device recommendation without server-side interaction logs, federated settings where data cannot leave the user’s

device, and cold-start contexts where population-level interaction graphs do not yet exist. The peer-isolation property is both a technical design choice and a practical deployment advantage.

7 Limitations and Future Work

- **Scale.** Last.fm-1K evaluation uses 99 users. While Amazon Books replicates on 2,000 users, larger-scale evaluation on both datasets would strengthen generalisability claims.
- **K sensitivity.** $K = 3$ is used throughout. An ablation over $K \in \{2, 3, 4, 5\}$ would characterise sensitivity to this hyperparameter across both domains.
- **Active cluster selection.** The recency-based heuristic is simple by design. A learned selection mechanism could improve session-aware accuracy without requiring cross-user training.
- **Filter bubble risk.** Peer-isolation removes the serendipity that cross-user signals can provide. Whether repeated centroid queries cause semantic narrowing is an open empirical question.
- **Embedding input curation.** Section 5.3 demonstrates that noisy embedding inputs degrade performance. The quality of what enters the embedding determines the quality of what Essence can find. Automated curation of embedding input text — filtering off-topic or irrelevant content before encoding — is a direct practical extension.
- **Audio-based embeddings.** Current embeddings derive from item metadata text only. For music, incorporating audio-based representations — capturing sonic texture, timbre, rhythm, and harmonic structure directly from audio signal — would more precisely model the latent acoustic fingerprint underlying individual taste. This represents the primary planned extension of Essence: replacing or augmenting text embeddings with audio embeddings and measuring whether long-tail recall improves when the embedding captures what the music actually sounds like rather than what it is called.
- **Cross-space embedding alignment.** Pass 2 results suggest that aligning review-text and metadata embedding spaces — via contrastive training or a learned projection layer — could unlock the full signal of user-authored representations while maintaining retrieval coherence.

8 Conclusion

We presented Essence, a personal embedding cluster framework for depth-optimised recommendations. Essence clusters a user’s item history into K semantic centroids and generates recommendations by cosine similarity to the currently active centroid, without consulting any cross-user data. Across two datasets — Last.fm-1K (music) and Amazon Books (e-commerce) — Essence consistently outperforms both popularity-based CF and content-based average embedding baselines on long-tail item recovery. CF achieves zero long-tail recall in every experimental condition. Essence achieves $4.5\times$ above the content baseline on Last.fm-1K and $1.28\times$ on Amazon Books, in both cases at $140\times$ or more above random.

The peer-isolation property is not merely a privacy safeguard — it is a depth requirement. Crowd signals approximate preferences at the genre level. Peer-isolation enables modelling at the latent feature level, recovering the idiosyncratic taste that collaborative signals structurally cannot reach. An artist discovered by accident under current infrastructure would be discovered by design under Essence: the latent signature of individual taste, encoded in a user’s own history, is sufficient to find it.

The long-tail metric generalises across datasets by anchoring the threshold to each dataset’s

interaction density — the point at which collaborative signal becomes unreliable — making LT-Recall@10 a portable evaluation instrument for niche item discovery across recommendation domains.

The primary contribution of this work is not a new embedding technique or a new clustering algorithm. It is the demonstration that *a single user’s own history contains sufficient signal to recover the depth of their taste* — without the crowd, without training, and without infrastructure. Discovery need not be accidental.

References

- [1] Cen, Y., et al. (2019). *MIND: Multi-Interest Network with Dynamic Routing for Recommendation at Tmall*. RecSys 2019.
- [2] Cen, Y., et al. (2020). *ComiRec: Controllable Multi-Interest Framework for Recommendation*. KDD 2020.
- [3] Sun, F., et al. (2019). *BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer*. CIKM 2019.
- [4] He, X., et al. (2017). *Neural Collaborative Filtering*. WWW 2017.
- [5] Koren, Y., Bell, R., & Volinsky, C. (2009). *Matrix Factorization Techniques for Recommender Systems*. IEEE Computer.
- [6] Hidasi, B., et al. (2016). *Session-Based Recommendations with Recurrent Neural Networks*. ICLR 2016.
- [7] Wang, W., et al. (2020). *MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers*. NeurIPS 2020.
- [8] Celma, O. (2010). *Music Recommendation and Discovery*. Springer.
- [9] Hou, Y., et al. (2024). *Bridging Language and Items for Retrieval and Recommendation*. arXiv:2403.03952.
- [10] Krichene, W., & Rendle, S. (2020). *On Sampled Metrics for Item Recommendation*. KDD 2020.